7 Conclusion

Section 6 establishes what the square example can tell us about both elastic deformation and the numerical work needed to compute it. Under rightward traction, the clamped square extends and contracts sideways. Its mean right-edge horizontal displacement approaches about 0.494 mm on the tested meshes, with decreasing changes under refinement. This supports a less mesh-dependent displacement prediction, but does not yet answer how different geometries or support arrangements change the response. Those comparisons remain proposed in Section 5.

For the supported symmetric positive-definite system, conjugate gradients is a practical starting choice among the tested iterative methods. It reaches the required residual with much less work than Gauss–Seidel in the recorded square benchmark. Diagonal preconditioning reduces the CG iteration count slightly, but does not improve the measured time, so it is not an automatic improvement. Ordinary Jacobi diverges, while damped Jacobi and steepest descent fail to reach the target within the iteration budget. These findings support checking convergence and total cost rather than selecting a method by simplicity or iteration count alone. The untimed sparse-LU solution remains a useful same-mesh reference, not a measured performance competitor.

The validation checks answer different questions. Prescribed displacements, force and moment balance, and agreement with the direct reference support the discrete solution. Refinement tests its dependence on the mesh; the separate exact affine test checks a different, roller-supported case. Neither establishes the continuum error of the clamped square. Moreover, conditioning worsens as the mesh is refined even while the measured displacements settle. The load-perturbation tests show that sensitivity also depends on how the load changes: the condition number describes possible amplification, not the response to every perturbation.

The present conclusions are therefore limited to the tested square and the small-deformation, plane-stress model. The next step is to agree and execute Section 5's geometry, load-location and support comparisons, preserving the physical loading and support regions under refinement. Each needs checked reference solutions before physical interpretation or solver recommendations can be extended. Permission to use the team's assembly routine must also be resolved, and physical validity near sharp corners needs further assessment. Completing these checks would allow the report to answer its original geometry-and-support question beyond one controlled example.